Ink Flow

Aspiring writers and hobbyists often struggle to share their thoughts online because modern website builders are overly complicated and crowded with unnecessary features.

A simple blog web app solves this by providing a streamlined, distraction-free platform where anyone can instantly write, edit, and publish posts without any technical expertise.

Who uses this?

• The Content Creator (The Writer): A person who wants to share thoughts, tutorials, or updates without dealing with a messy interface. They log in to their dashboard to write text, hit publish, and manage their archive.

• The Daily Reader (The Audience): Anyone arriving from a link or a search engine. They don't want to sign up; they just want a clean, fast-loading page to read the articles, browse categories, and leave a quick comment if they have something to say.

What does success look like?

Success means having a fully functional, zero-fluff web app where a writer can publish an article, and a reader can instantly see it.

- **The Happy Path Works:** A writer logs in, fills out a form with a title and content, hits "Publish," and the new post immediately appears at the top of the public homepage.
- **Clean and Fast Interfaces:** The writer has a distraction-free screen to manage their posts, and the reader gets a clean, fast-loading article view without any broken links or layout shifting.
- **Robust Data Handling:** When the server restarts, your data doesn't vanish. The articles and user accounts remain safely stored in the database.
- **Predictable Error Handling:** If a reader types in a URL for a blog post that doesn't exist, the app handles it gracefully with a clean "Post Not Found" page instead of crashing or showing a messy raw error code.

Blog Web App - Domains

S.No	Feature Name	Description / Functionality
1	User Authentication	Allows the writer to log in and securely access the dashboard.
2	User Management	Stores and manages writer accounts and user details.
3	Blog Post Management	Enables the writer to create, edit, publish, and delete blog posts.
4	Public Blog / Homepage	Displays published blog posts, with the newest posts appearing first.
5	Article Viewing	Allows readers to open and read individual blog posts without signing up.
6	Category Management	Organizes articles into categories for easier browsing.
7	Comment Management	Allows readers to leave comments on published articles.
8	Database Management	Persistently stores user accounts, blog posts, categories, and comments.
9	Error Handling	Displays friendly error pages, such as "Post Not Found," when a requested article does not exist.
10	Dashboard / Content Administration	Provides the writer with a simple interface to manage their published and draft posts.

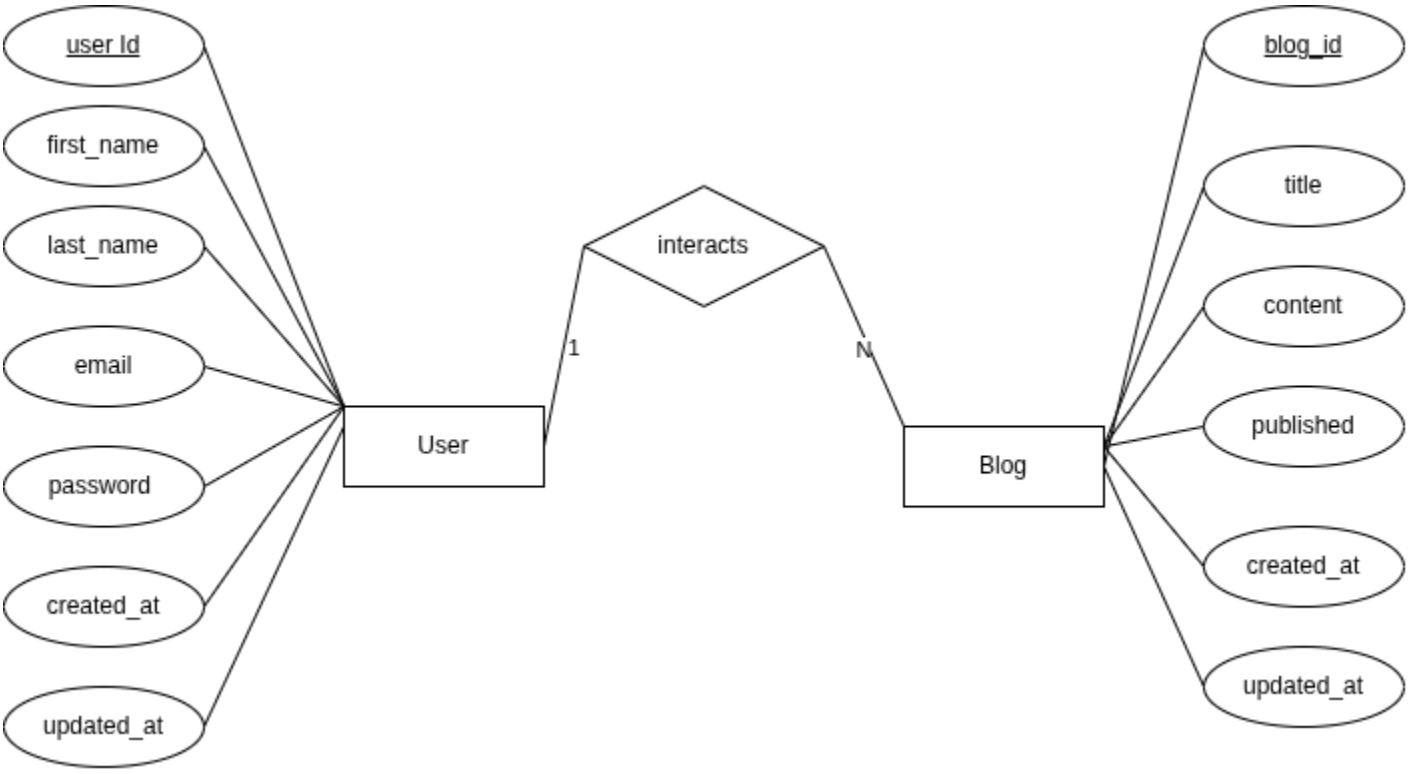
Core Domains

1. Authentication Writer login, logout, and protected dashboard access.
2. Blog Posts Create, edit, publish, delete, and retrieve articles.
3. Public Reading Homepage, individual article pages, and post-not-found handling.
4. Database and Persistence Permanent storage of accounts and articles across server restarts.
5. Writer Dashboard A distraction-free interface for managing drafts and published posts.

Ink Flow - MVP 1 Feature Set

1. **User Authentication:** Authors must be able to securely log in to their accounts and log out.
2. **Blog Management (CRUD):** Authors must be able to create, read, update, and delete their blog posts.
3. **Public Blog Access:** Readers must be able to view published blog posts without logging in. The application must display a clear "Post Not Found" page when a requested blog post does not exist.
4. **SQL Database Persistence:** Blog posts and user details must be persistently stored in a SQL database and remain available after application restarts.
5. **Simple and Clean UI:** The user interface must be minimal, responsive, and distraction-free for both authors and readers.
6. **Authorization and Access Control:** Only authenticated and authorized authors must be allowed to manage their own blog posts. Public readers must have read-only access to published posts.

MVP database ER diagram



Authentication APIs

Base path: /api/v1

Method	Endpoint	Access	Purpose
POST	/api/v1/signup	Public	Register a new author account
POST	/api/v1/signin	Public	Authenticate an author
POST	/api/v1/signout	Authenticated	Sign out the current author

Blog APIs

Method	Endpoint	Access	Purpose
POST	/api/v1/blogs	Author	Create a new blog post
GET	/api/v1/blogs	Public	List published blog posts
GET	/api/v1/blogs/{blogId}	Public	Read a published blog post by ID
PUT	/api/v1/blogs/{blogId}	Blog owner	Update an existing blog post
DELETE	/api/v1/blogs/{blogId}	Blog owner	Delete a blog post
PATCH	/api/v1/blogs/{blogId}/publish	Blog owner	Publish a draft
PATCH	/api/v1/blogs/{blogId}/conceal	Blog owner	Hide a published post or return it to draft status

Important authorization rules

Public readers: Can view only published blogs, without signing in.

Authenticated authors: Can create blogs and manage their own posts.

Blog ownership: An author cannot update, delete, publish, or conceal another author's post.

Draft visibility: Drafts must not be accessible through public read endpoints.

Not found: Return 404 Not Found for nonexistent posts. For private drafts, choose a consistent policy that avoids exposing their existence to unauthorized users.